Introduction to Radiomics and Machine Learning for Imaging

Instructor: Caryn Geady, PhD
Date: 9th March 2026

Today’s Objectives

- Understand the fundamentals of medical imaging and radiomics.
- Learn key ML techniques used in imaging-based predictive modeling.
- Recognize challenges and best practices in radiomics research.
- Briefly explore deep learning approaches in medical imaging.

https://www.stjoes.ca/hospital-services/diagnostic-services/diagnostic-imaging/requesting-your-images

Fundamentals of Medical Imaging

Images: more than pictures

- A matrix of numerical values;
- Pixel values have physical meaning;

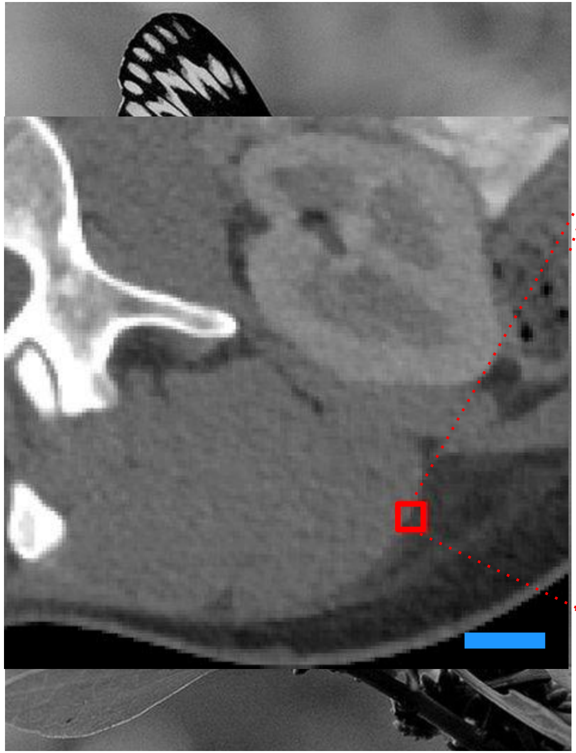
69	8	74	134	114
115	44	32	103	133
129	103	18	49	137
109	121	48	24	103
94	125	107	22	38

$Im \in \mathbb{R}^{m \times n}$

Photo: "Black and White" Collection - S. Paul

Images: more than pictures

- A matrix of numerical values;
- Pixel values have physical meaning;



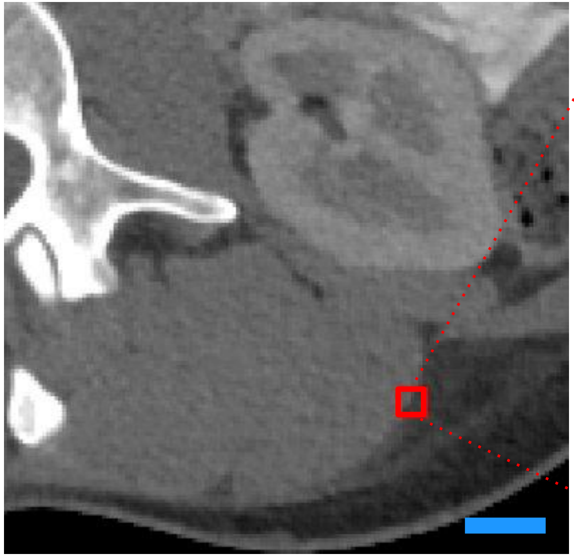
36	43	-1	-49	-60
38	48	-22	-44	-50
29	19	-34	-35	-29
1	-21	-46	-48	-31
-46	-45	-33	-43	-47

Scale bar represents a length of 2 cm

Photo: “Black and White” Collection – S. Paul

Images: more than pictures

- A matrix of numerical values;
- Pixel values have physical meaning;
- May encode clinically relevant information!



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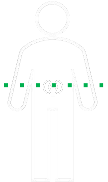
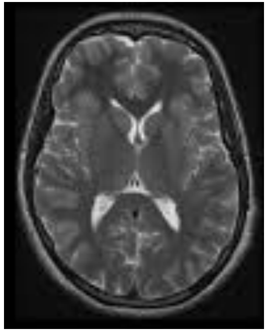


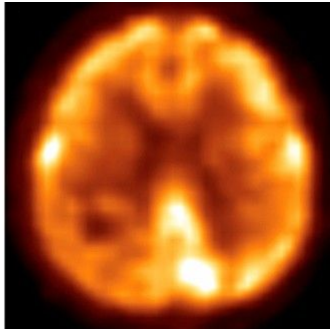
Scale bar represents a length of 2 cm

Medical Images

Computed Tomography (CT)



Magnetic Resonance (MR)



Positron Emission Tomography (PET)

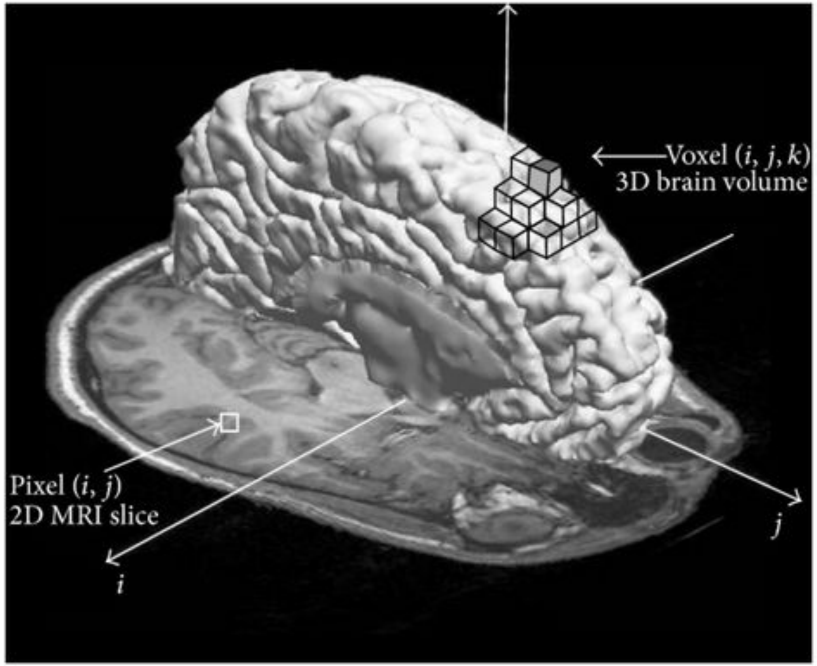
X-Ray

Ultrasound (US)

Medical Imaging Structure

Voxel Intensities

- **Definition:** A voxel (volumetric pixel) represents a 3D unit in medical imaging, with an assigned intensity value.
- **Modality Dependence:**
    - CT: Hounsfield Units (HU) indicate tissue density.
    - MRI: Signal intensity varies based on tissue properties and imaging sequences.
    - PET: Represents metabolic activity via standardized uptake values (SUV).



Interactive/Scrollable CT

https://onlinelibrary.wiley.com/doi/10.1155/2015/450341

Medical Imaging Structure

Metadata

- **Definition:** Additional information stored with the image, often in DICOM (Digital Imaging and Communications in Medicine) format.
- **Key Components:**
    - Patient details (ID, age, sex)
    - Acquisition parameters (modality, field strength, contrast use)
    - Image orientation and dimensions
    - Timestamp and institution information

(0018,0080)	DS	Repetition Time
(0018,0081)	DS	Echo Time
(0018,0082)	DS	Inversion Time
(0018,0083)	DS	Number of Averages
(0018,0084)	DS	Imaging Frequency
(0018,0085)	SH	Imaged Nucleus
(0018,0086)	IS	Echo Number(s)
(0018,0087)	DS	Magnetic Field Strength
(0018,0088)	DS	Spacing Between Slices
(0018,0089)	IS	Number of Phase Encoding Steps
(0018,0090)	DS	Data Collection Diameter
(0018,0091)	IS	Echo Train Length
(0018,0093)	DS	Percent Sampling
(0018,0094)	DS	Percent Phase Field of View
(0018,0095)	DS	Pixel Bandwidth
(0018,1000)	LO	Device Serial Number
(0018,1002)	UI	Device UID
(0018,1003)	LO	Device ID
(0018,1004)	LO	Plate ID
(0018,1005)	LO	Generator ID
(0018,1006)	LO	Grid ID
(0018,1007)	LO	Cassette ID
(0018,1008)	LO	Gantry ID

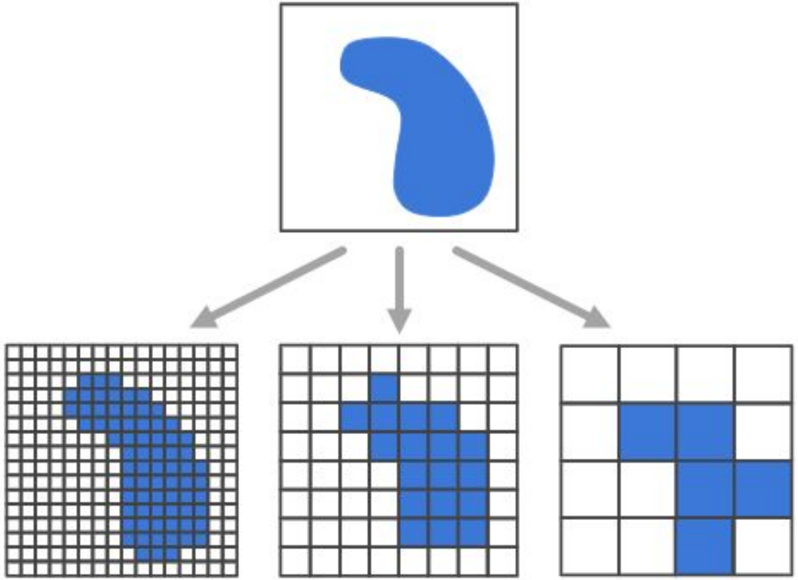
https://www.dicomlibrary.com/dicom/dicom-tags/

Interactive/Scrollable CT

Medical Imaging Structure

Spatial Resolution

- **Definition:** The ability to distinguish small structures in an image, measured in pixels per unit length (e.g., mm per pixel).
- **Factors Affecting Resolution:**
  - Voxel size: Smaller voxels ➔ Higher resolution
  - Modality Differences:
    - CT/MRI: Based on acquisition settings.
    - US: Depends on probe frequency.
    - X-ray: Limited by detector resolution.



[Interactive/Scrollable CT]()

https://learn.arcgis.com/en/projects/explore-imagery-spatial-resolution/

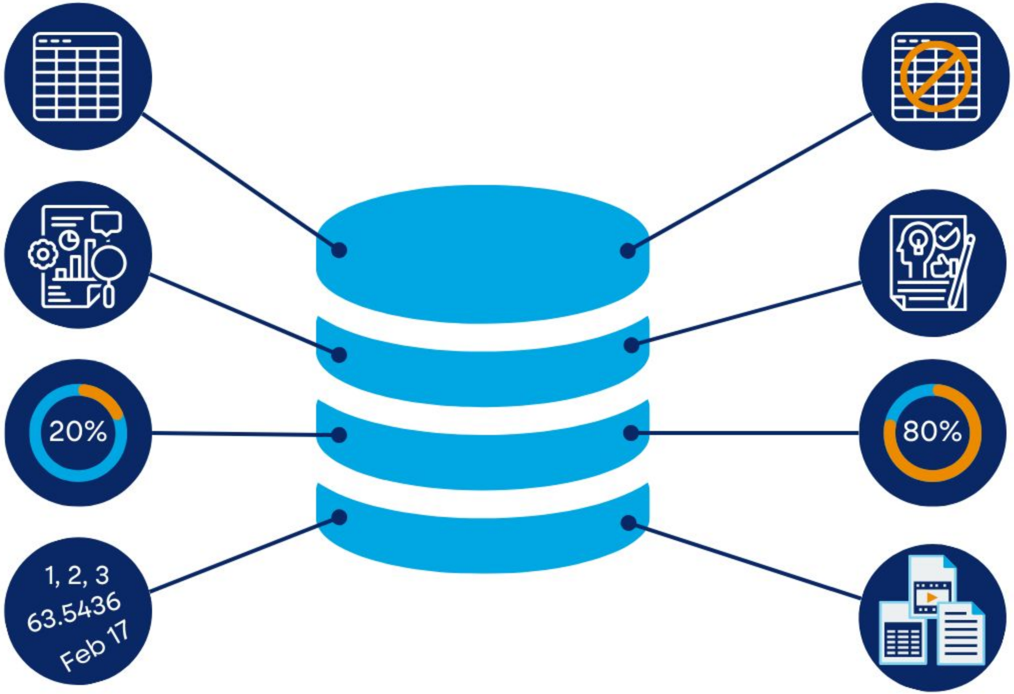
Structured versus Unstructured Data

Can be displayed in rows, columns, and relational databases

Quantitative

Estimated 20% of enterprise data (Gartner)

Numbers, dates, and strings



Cannot be displayed in rows, columns, or relational databases

Qualitative

Estimated 80% of enterprise data (Gartner)

Images, audio, video, word processing files, e-mails, spreadsheets

AI/ML in Medical Imaging: The Role of Data Structure

Structured Data in AI/ML

- Easier to integrate with traditional ML models.
- Supports clinical decision support systems (CDSS), patient stratification, and outcome prediction.

Unstructured Data in AI/ML

- Requires feature extraction techniques, i.e., radiomics, or deep learning (CNNs for images, NLP for text).
- Enables automated tumor detection, segmentation, and anomaly detection.

Medical Imaging Data is VERY Messy

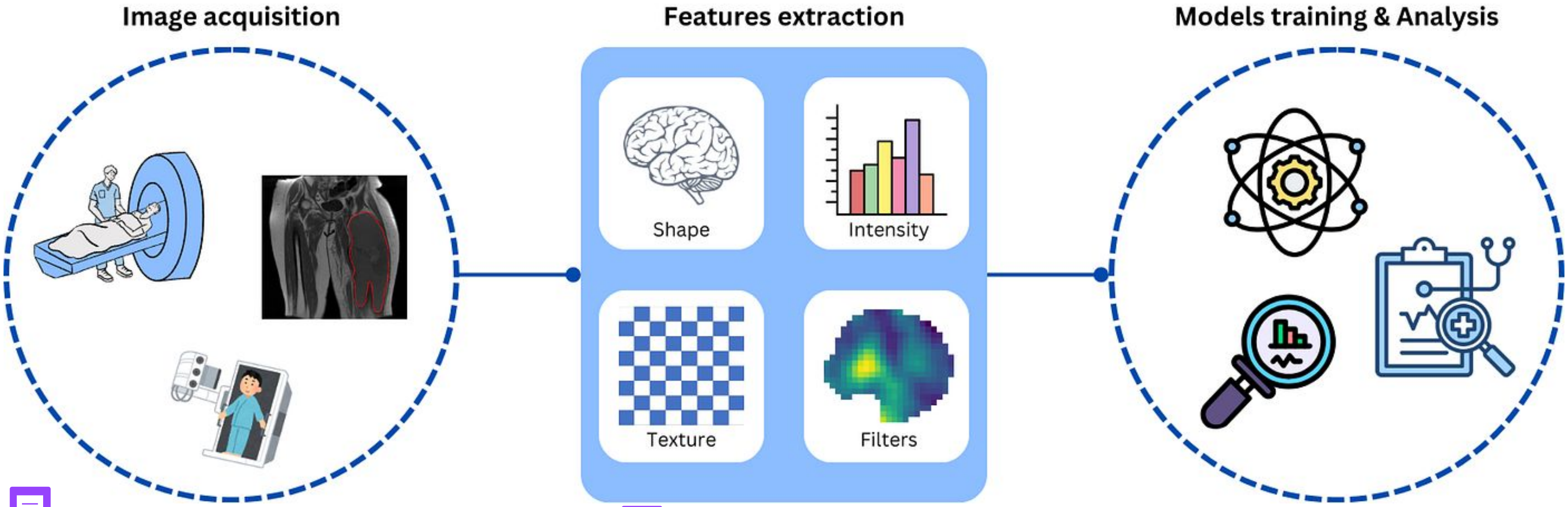
- **High variability** due to different scanners, protocols, and noise.
- Preprocessing is essential for AI/ML consistency and performance:
    a. Normalization: Adjusts intensity values to a common scale (e.g., Z-score, min-max scaling);
    b. Standardization: Ensures uniform voxel spacing, contrast, and resolution;
    c. Segmentation:

Method	Description	Pros	Cons
Manual	Expert-driven	Gold standard	Time-consuming, subjective
Semi-Automated	AI-assisted with user input	Faster, balances accuracy & efficiency	Still requires human intervention
Automated	Fully AI-driven (e.g., CNNs, U-Nets)	Fast, scalable, reproducible	May lack precision, requires validation

Introduction to Radiomics

What is Radiomics?

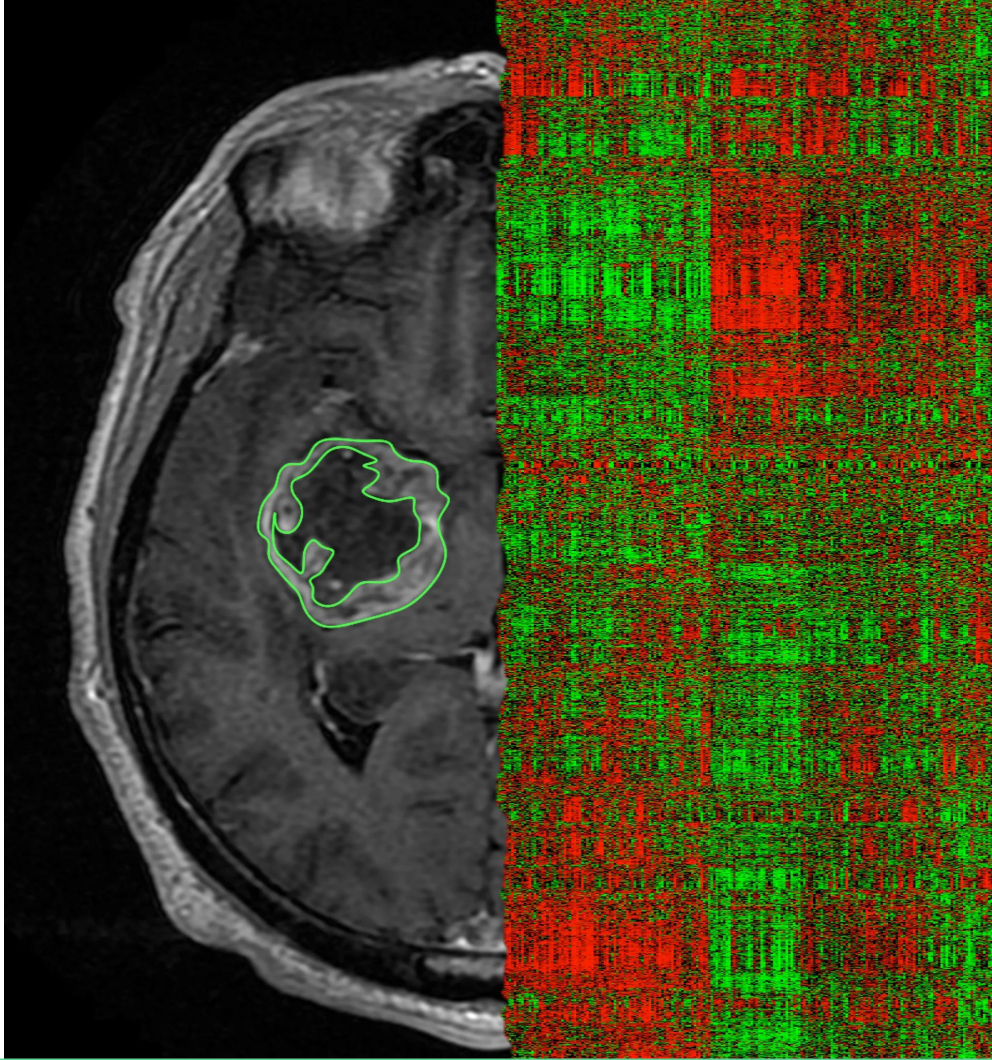
- Extraction of quantitative imaging features to uncover hidden patterns.



https://medium.com/@mahdiall99/radiomics-in-medical-imaging-a-brief-introduction-3208863aae87

Applications of Radiomics

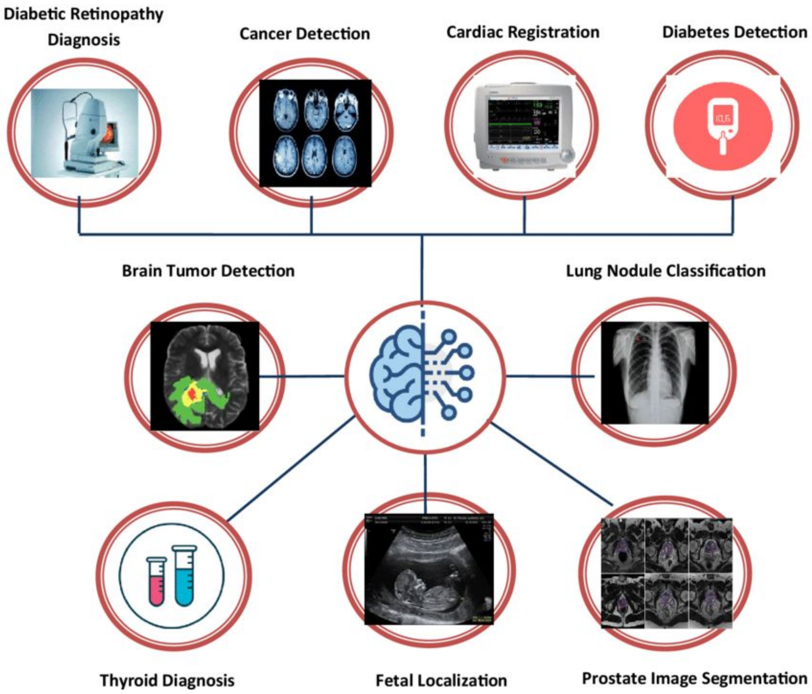
- **Predictive modeling:** Tumor progression, therapy response, survival;
- **Classification:** benign versus malignant, staging;
- **Risk stratification:** Identifying high-risk patients;
- **Multi-omics integration:** Combining radiomics with genomics & clinical data.



Machine Learning for Imaging

ML Techniques for Radiomics

- **Feature selection methods:**
  - LASSO,
  - PCA,
  - Clustering
  - ... and more!
- **Supervised learning approaches:**
  - Decision Trees,
  - Random Forest,
  - SVM,
  - Logistic Regression,
  - ... and more!



- **Evaluation Metrics:** AUC-ROC, precision-recall, Cross-validation, ... and more!

https://www.researchgate.net/figure/arious-Deep-learning-applications-in-Medical-Image-Processing_fig2_355417076

Introduction to Deep Learning for Imaging

When to Use Radiomics vs. Deep Learning

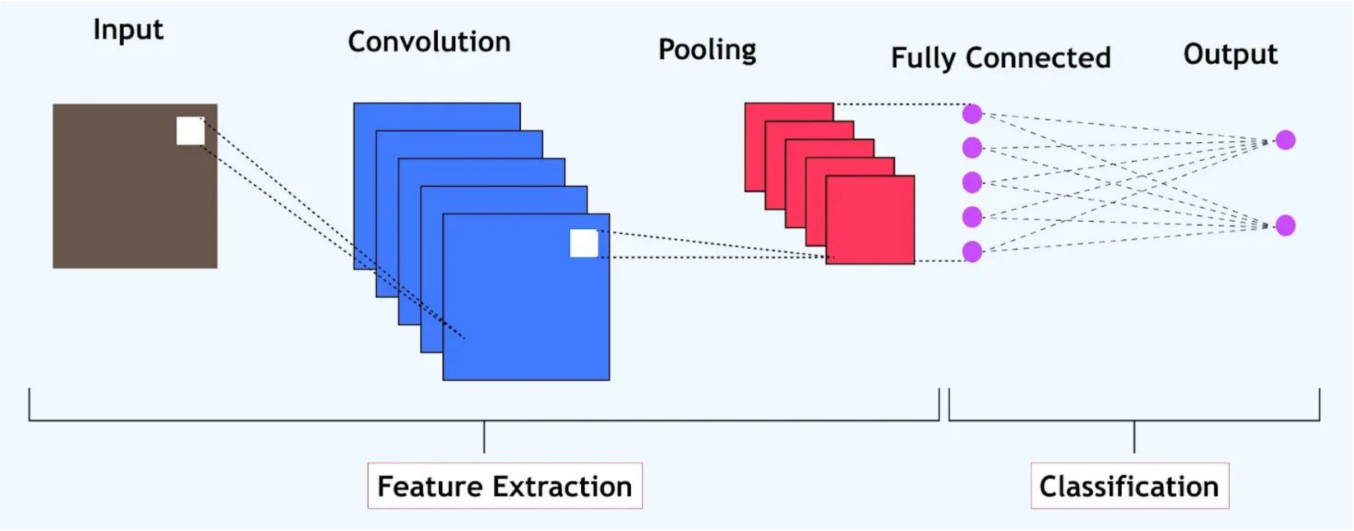
- Radiomics vs. deep learning:

Method	Strengths	Limitations
Radiomics	Small datasets, interpretable	Requires handcrafted features
Deep Learning (CNNs)	Fully automated, learns representations	Needs large labeled datasets

- Use cases for CNNs in medical imaging: Tumor classification, organ segmentation.

Overview of CNNs for Medical Imaging

- **Basic CNN architecture ➔** convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification.



Overview of CNNs for Medical Imaging

- **Basic CNN architecture ➔** convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification.
- **Why CNNs for medical imaging?** Effective for spatial patterns in images, commonly used for tasks like tumor detection, organ segmentation, and anomaly classification.
- **Transfer learning:** Pretrained models (e.g., ResNet, VGG) are useful when datasets are small, reducing the need for extensive labeled data.

Challenges

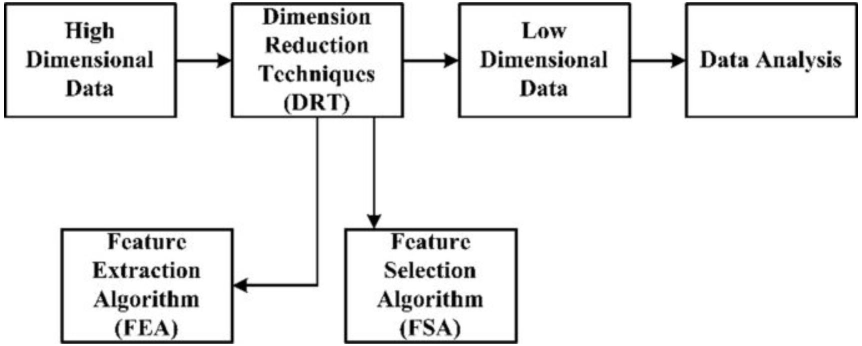
High Dimensionality and Overfitting

The Curse of Dimensionality

- Radiomics datasets are often small but high-dimensional, meaning they contain **many features but few samples**.
- High dimensionality increases the **risk of overfitting**;
- Overfitting leads to **poor generalizability** to new, unseen data.

Solution: Feature Reduction & Validation

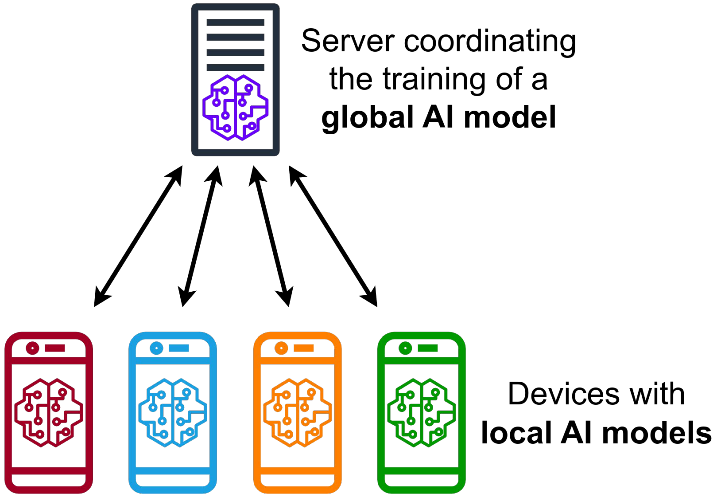
- L1 Regularization
- Dimensionality Reduction (i.e., PCA)
- K-Fold Cross-Validation



https://link.springer.com/article/10.1007/s10462-020-09928-0

Data Availability and Standardization

- Limited access to large, labeled datasets:
  - Medical imaging data is often fragmented across institutions.
  - Labeling requires expert annotations, which are time-consuming and expensive.
  - Privacy regulations (e.g., HIPAA, GDPR) limit data sharing.
- Public repositories for imaging data, e.g. TCIA (The Cancer Imaging Archive);
- Decentralized AI training without sharing raw patient data



https://en.wikipedia.org/wiki/Federated_learning

Reproducibility and Bias

Issue: Variability in imaging acquisition, segmentation methods

- Differences in image acquisition can introduce systematic variations;
- Manual vs. automated segmentation approaches;
- Lack of standardized preprocessing pipelines.

Solution: Open-access tools, standardized radiomics pipelines (IBSI)

- Image Biomarker Standardization Initiative (IBSI) for reproducible radiomics feature extraction;
- Public datasets (e.g., TCIA, BraTS, CheXpert) ensure diverse and representative training data;
- Federated learning approaches allow for multi-institutional collaboration;
- Data augmentation and domain adaptation techniques to mitigate scanner variability.

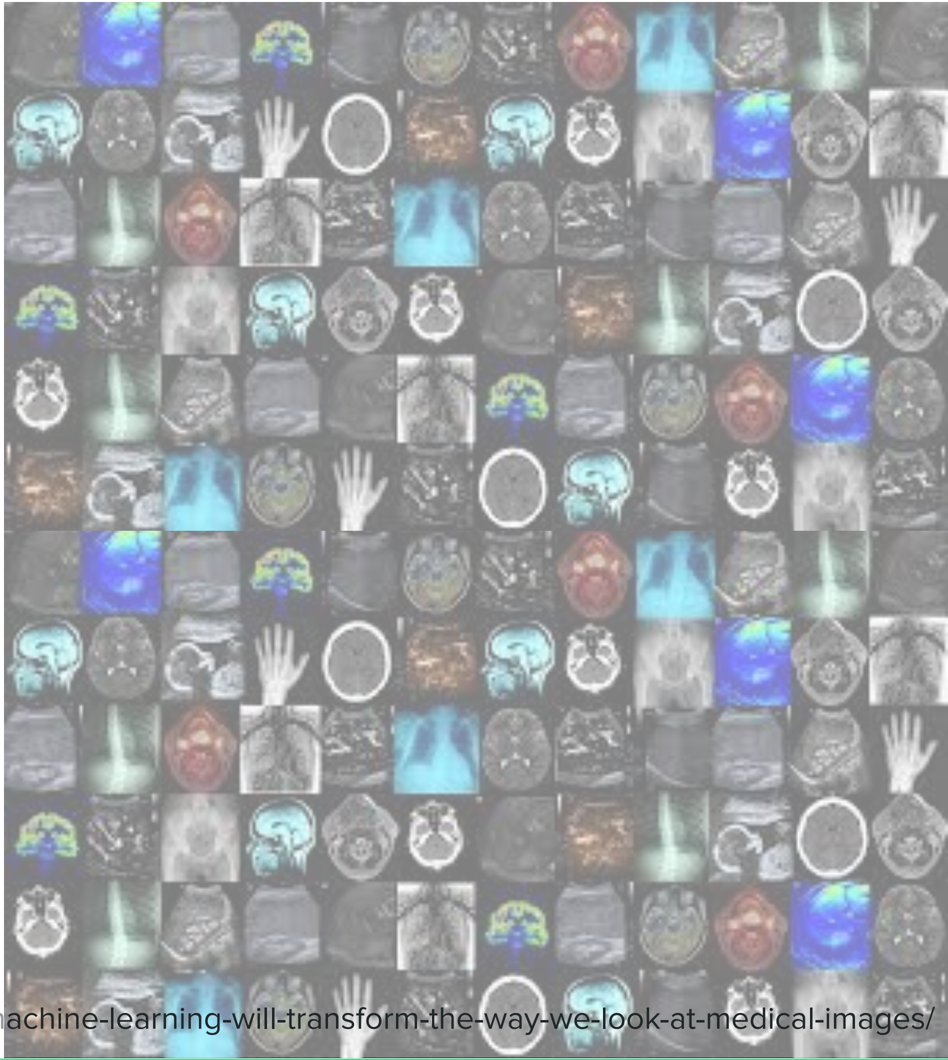
Summary & Discussion

Medical Imaging Modalities and Clinical Uses

Imaging Modality	Principle	Strengths	Limitations	Primary Clinical Use Cases
CT (Computed Tomography)	X-ray-based cross-sectional imaging	Fast, high-resolution, good for bone and soft tissue, can be enhanced with contrast	Radiation exposure, limited soft tissue contrast compared to MRI	Trauma (e.g., fractures, internal bleeding), stroke assessment, tumor detection, lung and abdominal imaging
MR (Magnetic Resonance)	Uses strong magnetic fields and radio waves	Excellent soft tissue contrast, no radiation, good for brain, muscles, and joints	Longer scan times, expensive, contraindicated in patients with metal implants	Neurological disorders (e.g., stroke, tumors), musculoskeletal injuries, spinal cord evaluation, soft tissue tumors
PET (Positron Emission Tomography)	Uses radioactive tracers to assess metabolic activity	Functional imaging, good for detecting cancer, monitoring treatment response	Radiation exposure, lower spatial resolution, often combined with CT (PET-CT)	Cancer staging and response monitoring, neurological disorders (e.g., Alzheimer's), cardiac perfusion imaging
US (Ultrasound)	High-frequency sound waves	No radiation, real-time imaging, portable, safe for pregnancy	Operator-dependent, limited penetration in obese patients and gas-filled structures	Obstetrics (fetal imaging), abdominal and cardiac imaging, vascular studies, musculoskeletal assessment
X-ray	Uses ionizing radiation to create 2D images	Fast, inexpensive, good for bone imaging	Radiation exposure, limited soft tissue contrast	Fractures, chest imaging (pneumonia, lung disease), dental imaging

Key Takeaways

- Radiomics extracts quantitative imaging features for ML models;
- ML techniques (Random Forest, SVM, LASSO) work well for structured imaging data;
- Deep learning (CNNs) is powerful but requires large datasets and compute power;
- Challenges include small datasets, overfitting, and reproducibility.



https://blogs.imperial.ac.uk/imperial-medicine/2018/01/17/how-machine-learning-will-transform-the-way-we-look-at-medical-images/

Questions / Food for Thought

1. How can radiomics be integrated into clinical workflows?
2. What strategies can improve reproducibility in AI-driven radiomics?
3. How do we balance interpretability and performance in AI models for imaging?
4. ...?

https://www.redbubble.com/shop/cat+scan+posters